

# MUHAMMAD HAZIQ HASBI BIN OSMAN

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## PROFESSIONAL SUMMARY

Technically oriented electronic engineering graduate with hands-on exposure to web development, electronics, semiconductor design verification, IoT and machine learning through internships, technical training and academic projects. Experienced in developing web-based workflow systems using HTML, CSS and JavaScript, with additional technical knowledge of SystemVerilog, Verilog, UVM, Linux, C++, Python and MATLAB. Practical experience includes PCB soldering, component replacement, troubleshooting, preventive maintenance and machine buy-off, supported by exposure to semiconductor manufacturing and network infrastructure. Demonstrates strong problem-solving, analytical thinking, adaptability and teamwork, with a strong interest in contributing to technology-driven and engineering environments.

## WORK EXPERIENCE

### On-The-Job Training (OJT) – K-Youth Development Programme Fortuntech Intelligence Sdn. Bhd

June 2026 – Present

- Developing a web-based Office Approval System to digitalise and streamline internal employee approval workflows and administrative processes.
- Designing and developing responsive web interfaces using HTML, CSS and JavaScript, focusing on usability, accessibility and efficient navigation.
- Implementing role-based approval workflows for Employees, Supervisors, Managers/HODs, HR/Admin and Administrators to support structured request processing.
- Developing digital modules for Overtime, Leave, Loan, Travel and General Approval requests, replacing manual processes with centralised online submissions.
- Building approval dashboards, request status tracking and automated routing functions to improve workflow visibility and facilitate timely decision-making.
- Conducting system testing, debugging and continuous enhancements based on business requirements and user feedback to improve functionality and reliability.

### K-Youth Development Programme Khazanah Nasional Aisling Consulting Sdn. Bhd

May 2026 – June 2026

- Completed workforce readiness and professional development training covering professional communication, workplace adaptability, teamwork and career preparedness.
- Strengthened workplace competencies through structured training designed to support effective transition into a professional working environment.

### IC Design Training Program: Design Verification Penang Skill Development Center (PSDC)

Nov 2025 – Dec 2025

- Developed and executed SystemVerilog and Verilog testbenches for digital IC verification, including simulation execution, waveform analysis and debugging in a Linux-based environment.
- Applied fundamental digital IC design and CMOS concepts, including transistor operation, logic gates and their application in digital circuit design.
- Gained practical knowledge of UVM (Universal Verification Methodology) and verification flows used in functional verification.
- Developed basic UVM components including sequences, drivers, monitors and scoreboards to support structured verification environments.
- Practiced SystemVerilog Assertions (SVA) to verify functional behaviour, timing conditions and design requirements.
- Applied simulation-based and assertion-based verification techniques through practical laboratory exercises and debugging activities.

### Degree Internship

July 2024 – Sept 2024

#### Robert Bosch Semiconductor manufacturing Penang Sdn. Bhd

- Developed an Excel-based spare parts tracking system using structured tables to improve inventory visibility, tracking and record management.
- Conducted testing and repair of workpiece carriers, identifying defective components and performing root cause analysis to support equipment functionality.
- Performed soldering and component replacement on PCB assemblies in accordance with ESD and IPC requirements for quality rework.
- Supported troubleshooting and technical inspection activities to identify component-level issues and restore workpiece carrier functionality.
- Maintained accurate technical and inventory-related records to support spare parts tracking and maintenance activities.
- Applied practical knowledge of electronics, PCB rework, troubleshooting and manufacturing processes within a semiconductor production environment.

### Diploma Internship

Mar 2021 – July 2021

#### SSIS E Solution Sdn. Bhd

- Assisted with the installation and configuration of fiber optic networks and server racks, following network layouts and

workplace safety requirements.

- Performed Ethernet cable termination, routing and labelling for LAN infrastructure to support reliable network connectivity.
- Supported the installation and configuration of network switches, routers and patch panels for client infrastructure projects.
- Conducted signal testing, troubleshooting and cable certification using network diagnostic tools to verify network link performance.
- Assisted with network infrastructure deployment and ensured cabling and equipment were properly organised according to project requirements.
- Maintained installation records and network layout documentation to support project handover, traceability and compliance requirements.

## EDUCATION

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### Bachelor of Degree with Honours in Electronic Engineering

Sept 2021 – July 2025

Universiti Tun Hussein Onn Malaysia (UTHM)

Dean's List Award – Semester 2 & 7

### Diploma in Electrical & Electronic Engineering

June 2018 – Aug 2021

Politeknik Seberang Perai

## ACADEMIC PROJECTS

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### Maternal Health Risk Prediction using Tree-Based Machine Learning Models | Degree Final Year Project – 2025

- Developed a tree-based machine learning model to classify maternal health risk levels as low, medium or high using clinical attributes including blood pressure, blood sugar and heart rate.
- Applied data preprocessing, feature analysis and classification techniques to develop and evaluate the predictive model.

### IoT-Based Energy Consumption Monitoring System | Degree Integrated Design Project – 2024

#### Silver Award – Integrated Design Project (IDP) Showcase, UTHM

- Designed and developed an IoT-based system for real-time monitoring and analysis of energy consumption in residential and small-scale industrial environments.
- Integrated sensors and remote monitoring capabilities to support energy usage optimisation, consumption analysis and cost reduction.

### Autonomous Fire-Fighting Robot | Diploma Final Year Project – 2021

- Designed and developed an autonomous fire-fighting robot capable of detecting fire sources, navigating toward them and activating a suppression mechanism.
- Integrated sensors, robotics, electronics and control systems to enable autonomous fire detection and early-response capabilities.

## CERTIFICATIONS & TRAINING

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- **IC Design Training Program: Design Verification** – Penang Skill Development Centre (PSDC) | 2025
- **Training: MATLAB – Calculus & Ordinary Differential Equations** | 2024
- **SULAM Project** – Faculty of Electrical and Electronic Engineering (FKEE), UTHM | 2023
- **Production Workshop Program** – UTHM Multimedia Club | 2021

## SKILLS & COMPETENCIES

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### Technical Skills

- **Programming & Development:** C++, Python, HTML, CSS, JavaScript
- **Engineering & Simulation Software:** MATLAB, Multisim, Proteus, Arduino IDE, Vivado, Xilinx SDK
- **Systems & Productivity Tools:** Linux, Microsoft Office (Word, Excel, PowerPoint, Teams)
- **Electronics & Manufacturing:** PCB Soldering, Component Replacement, Assembly & Product Testing, Preventive Maintenance, Machine Buy-Off, Basic Troubleshooting
- **IC Design & Verification:** SystemVerilog, Verilog, UVM, SystemVerilog Assertions (SVA), Digital IC Design & Verification

### Soft Skills

- Problem-Solving & Analytical Thinking
- Technical Learning & Adaptability
- Teamwork & Collaboration
- Communication & Interpersonal Skill
- Time Management & Work Organisation

### Languages

- **Malay:** Native
- **English:** Professional Working Proficiency

## REFERENCES

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### Dr Nur Ilyani Binti Ramli

Supervisor Final Year Project

Universiti Tun Hussein Onn Malaysia (UTHM)

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### Anbarasu Balasubramaniam

Supervisor

Robert Bosch Semiconductor Manufacturing

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